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Low Cognitive Function and Somatic Psychological Symptoms Are Correlated with Greater Risk of Delirium After Total Knee Arthroplasty

A Prospective Cohort Study

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Background: Postoperative delirium (POD) is a clinically important complication in elderly patients undergoing total knee arthroplasty (TKA) that is associated with prolonged hospitalization, increased morbidity, and higher health-care costs. Although cognitive impairment is a known risk factor for POD, the role of comprehensive cognitive and psychological evaluation remains underexplored in patients undergoing TKA. This study aimed to evaluate the correlation of preoperative cognitive and psychological factors with POD after TKA.

Methods: This prospective cohort study included 574 patients who were ≥60 years of age and underwent primary TKA at 1 of 2 major tertiary care hospitals. We assessed preoperative cognitive function using the Mini-Mental State Examination (MMSE), the full Consortium to Establish a Registry for Alzheimer's Disease (CERAD) battery, the Subjective Memory Complaints Questionnaire (SMCQ), and the Seoul Informant Report Questionnaire for Dementia (SIRQD). Psychological assessments were conducted with the Pittsburgh Sleep Quality Index (PSQI), the Patient Health Questionnaire-15 (PHQ-15), and the Hospital Anxiety and Depression Scale (HADS). POD was evaluated daily from postoperative days 1 to 5 using the 4 A's Test (4AT) and the Confusion Assessment Method (CAM). A multivariable logistic regression analysis was performed to identify independent risk factors for POD.

Results: POD occurred in 24 (4.2%) of 574 patients. Univariate analysis revealed that POD was significantly correlated with lower MMSE ($p < 0.001$), higher PHQ-15 ($p = 0.014$), higher PSQI ($p = 0.014$), and higher Charlson Comorbidity Index ($p = 0.010$) scores; preoperative use of sedatives ($p = 0.044$) and antidepressants ($p = 0.027$); and lower mean noise levels in the patient's hospital room ($p = 0.002$). In the receiver operating characteristic curve analysis, the optimal cutoff value for predicting POD was an MMSE score of ≤25, with a sensitivity of 74.5% and a specificity of 78.3% (area under the curve, 0.834; $p = 0.001$). Multivariable logistic regression analysis identified lower MMSE scores (odds ratio [OR], 0.771; $p = 0.002$) and higher PHQ-15 scores (OR, 1.187; $p = 0.028$) as significant independent predictors of POD.

Conclusions: This study comprehensively evaluated preoperative cognitive function and psychological symptoms in patients undergoing TKA. Even subclinical cognitive and somatic symptoms were linked to POD, emphasizing the need for preoperative identification of high-risk patients.

Level of Evidence: Prognostic Level II. See Instructions for Authors for a complete description of levels of evidence.

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Postoperative delirium (POD) is a clinically important complication in elderly patients undergoing major orthopaedic surgeries, including total knee arthroplasty (TKA)[1-8]. The reported incidence of POD following TKA varies considerably, with rates ranging from 0% to 0.7% in fast-track arthroplasty registry data[4,9] to 3.1% to 6.5% in single-center cohorts[2,3] and up to 1.5% to 7.1% in studies involving total joint arthroplasty populations that include both TKA and total hip arthroplasty (THA)[1,5]. POD manifests as acute confusion, inattention, and fluctuating mental status, resulting in increased hospital stay length, morbidity, and health-care expenses[10,11]. Considering its impact on postoperative recovery and outcomes, the identification of modifiable risk factors for POD is necessary.

The primary findings of this prospective cohort study are that lower preoperative MMSE scores and higher PHQ-15 scores significantly predicted POD in patients undergoing TKA.

Although cognitive impairment is a well-established risk factor for POD, most studies have relied on brief assessments such as the Mini-Mental State Examination (MMSE), which fail to capture the full spectrum of cognitive functions[12,13]. Additionally, preoperative psychological symptoms such as sleep disturbances, anxiety, and depression have been linked to an increased risk of POD[14-16]. Given the high prevalence of these conditions among elderly patients undergoing TKA[17-20], a comprehensive assessment of cognitive and psychological functions is crucial for identifying modifiable preoperative risk factors and developing targeted interventions to prevent POD[1,13,21]. Recent systematic reviews have further underscored this need, emphasizing the multifactorial nature of POD and noting that most available evidence derives from heterogeneous surgical cohorts rather than TKA populations specifically[22,23].

Consequently, in this study we hypothesized that specific aspects of cognitive and psychological functions may increase the risk of POD in patients undergoing TKA. Additionally, it evaluated preoperative cognitive and psychological functions and their correlation with POD, while integrating a comprehensive spectrum of potential risk factors, such as comorbidities, laboratory findings, baseline medication usage, and environmental factors.

Materials and Methods

A prospective cohort study design was implemented to recruit patients undergoing primary TKA at 2 major tertiary care centers from May 2021 to December 2024. These facilities adhered to comparable hospitalization standards and operational techniques for patients who underwent TKA, enhancing consistency in perioperative management.

The hospitals' institutional review board approved the study (no. 2305-138-1434).

The study enrolled patients who were ≥60 years of age and were scheduled for primary TKA due to degenerative osteoarthritis, confirmed by a Kellgren-Lawrence grade of ≥3 on radiographic evaluation. Patients were excluded if they had a clinical diagnosis of dementia, were taking dementia-related medication, or were considered unsuitable for participation by the principal investigator.

The attending physicians reviewed each patient's medical history, explained the study to them, and obtained their informed consent 1 day prior to surgery. On the same day, research evaluators who had undergone standardized training from the principal investigator conducted cognitive and psychological assessments. These evaluators also collected demographic and clinical information, data on socioeconomic factors, laboratory results, and environmental conditions before surgery.

Cognitive function was evaluated using the Korean version of the Consortium to Establish a Registry for Alzheimer's Disease (CERAD) neuropsychological battery, which comprises 9 tests: J1 (Verbal Fluency), J2 (Boston Naming Test), J3 (MMSE), J4 (Wordlist Learning), J5 (Constructional Praxis), J6 (Wordlist Recall), J7 (Wordlist Recognition), J8 (Constructional Praxis Recall), and J9 (Trail Making Test, Parts A and B)[24,25]. The CERAD total II score was calculated by summing the J1, J2, J4, J5, J6, J7, and J8 scores. Subjective cognitive symptoms were assessed through the Subjective Memory Complaints Questionnaire (SMCQ), and caregiver-reported cognitive decline was evaluated using the Seoul Informant Report Questionnaire for Dementia (SIRQD)[26,27].

Psychological and somatic symptom assessments included the Hospital Anxiety and Depression Scale (HADS) to evaluate anxiety and depressive symptoms; the Pittsburgh Sleep Quality Index (PSQI) global score, which is the sum of 7 component scores, to assess overall sleep quality; and the Patient Health Questionnaire-15 (PHQ-15) to gauge the burden of somatic symptoms[28-30]. Preoperative pain intensity was assessed using the Numeric Rating Scale (NRS).

Demographic and clinical data included age, body mass index (BMI), sex, history of delirium, comorbidities (e.g., hypertension, diabetes mellitus, chronic kidney disease, cerebrovascular disease, and solid tumor), and the Charlson Comorbidity Index (CCI). The use of glasses or hearing aids was documented. Socioeconomic factors, including education level, employment status, medical coverage type, marital and parental status, history of alcohol use, and history of smoking, were documented.

Laboratory evaluations included hemoglobin, blood urea nitrogen (BUN), creatinine, estimated glomerular filtration rate (eGFR), albumin, sodium, potassium, erythrocyte sedimentation rate (ESR), and C-reactive protein (CRP)[3,7,31-35].

The use of preoperative medications, including opioids, sedatives, antidepressants, antipsychotics, antiepileptics, cholinergic agents, and anticholinergic agents, was also documented[36].

Regarding environmental conditions, noise levels were measured before and after the preoperative interview, and the mean of these levels was used for analysis. A sound level meter placed 1 m from the patient's bed recorded the noise in decibels (dB). Illuminance (lux) levels in the patient's hospital room and the bed's position relative to the window were recorded.

Postoperative evaluations were conducted daily from postoperative days 1 to 5 by trained research evaluators. All patients were screened using 2 standardized tools: the 4 A's Test (4AT: Arousal, Attention, Abbreviated Mental Test-4, Acute change) and the Confusion Assessment Method (CAM)[37,38]. POD was defined as a CAM-positive result (see Appendix Supplementary Table 1). The Delirium Rating Scale-revised-98 was administered to patients who scored positive on the 4AT but did not meet CAM criteria, in order to identify subsyndromal delirium[39].

Surgical data included American Society of Anesthesiologists (ASA) classification, anesthesia type (general or spinal), and intraoperative transfusion history. General anesthesia was administered only when spinal anesthesia was contraindicated, such as in patients taking antiplatelet medications that could not be discontinued, after consultation with the anesthesiology team.

Intraoperative analgesia was provided by periarticular injection administered before implantation[40]. Postoperative pain management followed a standardized multimodal analgesic protocol, consisting of scheduled non-opioid analgesics and a continuous adductor canal block using an indwelling catheter maintained until postoperative day 3, with opioids provided only as rescue medication when necessary.

Statistical Analysis

Differences in continuous variables were assessed using the independent t test, and categorical variables were analyzed with the chi-square test. Variables demonstrating significant differences between the groups in univariate analyses were checked for multicollinearity, defined as a tolerance of <0.2 or variance inflation factor (VIF) of >5, and were excluded from the logistic regression analysis if multicollinearity was present[41]. Multivariable logistic regression analyses were conducted to identify significant risk factors for POD. Additionally, odds ratios (ORs), along with 95% confidence intervals (CIs), were calculated. Based on clinical relevance and consideration of multicollinearity, potential confounders (e.g., age, BMI, CCI, noise level, and pain NRS) were adjusted for in the logistic regression models[42]. Receiver operating characteristic (ROC) curve analyses were performed to determine the optimal cutoff values for MMSE, PSQI, and PHQ-15 scores in predicting POD. The Youden index was used to identify the point maximizing sensitivity and specificity, and the area under the curve (AUC) and corresponding p values were calculated. A 2-tailed p value of <0.05 was considered significant for all analyses. Statistical analyses were conducted using SPSS (version 31; IBM), and a post hoc power analysis was performed with G*Power (version 3.1.9.7; Heinrich-Heine-Universität Düsseldorf).

Results

Incidence of POD

POD occurred in 24 (4.2%) of 574 patients, as detailed in the patient flowchart (Fig. 1). Among these 24 patients, 17 experienced delirium on the day of surgery and 7 developed it on postoperative day 1, with onset at a mean of 0.29 ± 0.46 days.

Demographic and Baseline Characteristics of Groups with and without POD

The demographic and baseline characteristics of patients with and without POD are summarized in Table 1. The group with POD, compared with the group without POD, had a higher CCI (3.39 ± 2.21 compared with 1.96 ± 1.77; p = 0.010) and more frequent use of sedatives (16.7% compared with 6.5%; p = 0.044) and antidepressants (12.5% compared with 3.8%; p = 0.027). The mean noise level was lower in the group with POD (50.5 ± 7.4 compared with 56.4 ± 15.2 dB; p = 0.002).

Data Completeness of Cognitive and Psychological Assessments

Due to the self-reported nature of assessments and varying levels of patient compliance, missing values occurred and were excluded from the respective analyses (see Appendix Supplementary Table 2). The results of the CERAD subtests and psychological assessments are presented in Appendix Supplementary Table 3.

Comparison of Cognitive and Psychological Assessments Between Groups

Preoperative cognitive and psychological assessment scores differed significantly between the group with POD and the group without POD, as shown in Table 2.

The CERAD total II score was lower in the group with POD (39.26 ± 30.21 compared with 51.72 ± 33.21; p = 0.072), although this did not reach significance, and there were significant differences in its domains, including the MMSE (23.47 ± 3.18 compared with 26.28 ± 3.55; p < 0.001) (see Appendix Supplementary Figure 1).

Regarding the psychological factors, the POD group exhibited higher PSQI scores (11.05 ± 4.17 compared with 8.40 ± 3.93; p = 0.014) and PHQ-15 scores (5.92 ± 3.21 compared with 4.28 ± 2.57; p = 0.014).

In the ROC curve analysis, the optimal MMSE cutoff value for predicting POD was identified as a score of ≤25, with a sensitivity of 74.5% and a specificity of 78.3%. The AUC was 0.834 (p = 0.001), as shown in Appendix Supplementary Figure 2. However, the ROC analyses for the PSQI (AUC, 0.657; p = 0.059) and PHQ-15 (AUC, 0.633; p = 0.077) did not reach significance.

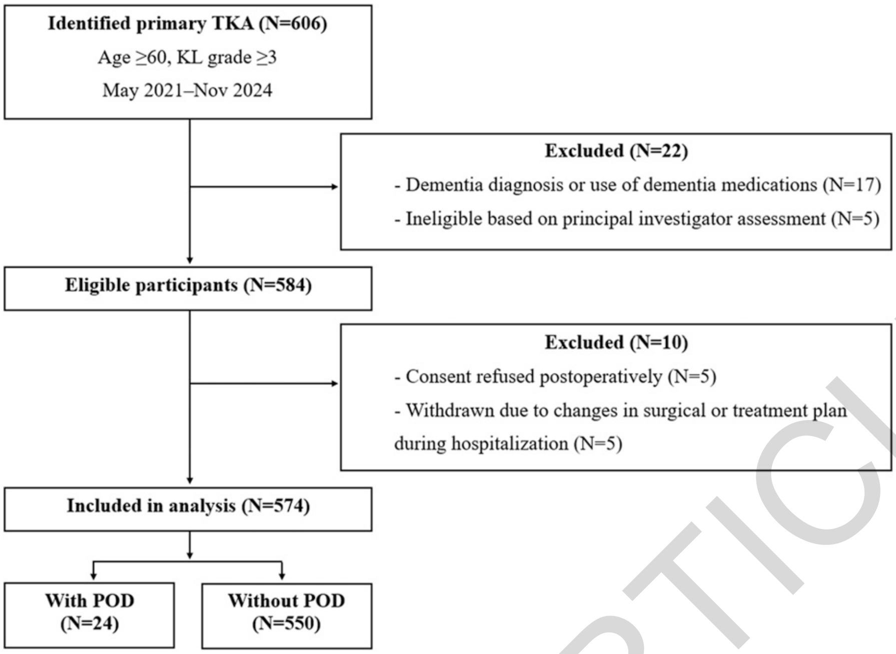

Fig. 1

Patient flowchart. KL = Kellgren-Lawrence.

Logistic Regression Analysis of POD Risk Factors

After adjusting for age, BMI, CCI, noise level, and pain NRS, both the MMSE (Table 3) and PHQ-15 (Table 4) were identified as significant predictors of POD (Tables 3 and 4), whereas PSQI did not reach significance (see Appendix Supplementary Table 4). Higher preoperative MMSE scores were significantly correlated with lower risk of POD (OR, 0.771; p = 0.002), and higher PHQ-15 scores were significantly correlated with higher risk (OR, 1.187; p = 0.028).

Discussion

The primary findings of this prospective cohort study are that lower preoperative MMSE scores and higher PHQ-15 scores significantly predicted POD in patients undergoing TKA. Moreover, the preoperative use of sedatives and antidepressant medications was more prevalent in the group with POD, and noise levels in the hospital environment were notably lower compared with the group without POD. Considering that an MMSE of ≤25 was correlated with an increased risk of POD, such patients should receive closer perioperative monitoring to mitigate the risk of POD.

The overall POD incidence herein was 4.2%, which is lower than those reported for hip fracture surgeries (19.3%), cardiac surgeries (20.1%), and bone metastasis surgeries (9.1%) (see Appendix Supplementary Table 5)[6,7,43]. Conversely, this rate is comparable with those from previous studies on POD in elective orthopaedic surgeries, such as TKA, THA, or spine surgery, which documented incidence rates ranging from 1.5% to 7.1%[1-5,8,9].

Several factors may explain the lower POD incidence observed in patients who undergo TKA compared with other surgical cohorts. The baseline health status of patients who undergo TKA substantially differs from that of patients undergoing hip fracture, cardiac, or bone metastasis surgeries[6,7,43]. Unlike patients undergoing hip fracture surgery or cardiac surgery, who must endure intensive care unit stays and prolonged immobilization, patients who undergo TKA are typically subjected to early mobilization and rehabilitation protocols that have proven to diminish the risk of POD[44,45].

Lower MMSE scores and higher PHQ-15 scores were identified as independent predictors of POD. The final logistic regression model included age, BMI, CCI, noise level, and pain NRS as potential confounders, along with the candidate variables MMSE, PSQI, and PHQ-15. Only the MMSE and PHQ-15 remained significant, suggesting that cognitive function and the burden of somatic symptoms, rather than sleep quality, more directly influence the risk of POD in patients who underwent TKA.

High background noise has been recognized as a potential contributor to delirium, particularly in intensive care unit settings, which can disrupt normal sleep-wake cycles[46-48]. However, our study revealed that the group with POD experienced lower preoperative noise levels compared with the group without POD. The World Health Organization-recommended maximum of 35 dB for patient rooms was exceeded in both groups, indicating exposure to relatively high noise levels[49]. After adjustment for covariates, noise level was not independently correlated with POD. Given that measurements were taken in the general ward on the day before surgery, the observed difference might reflect transient variations in environmental conditions rather than a protective or harmful effect of noise itself. We interpret this finding as exploratory and context-dependent, not causal, and recognize the potential for residual confounding related to the room context and measurement timing. The relationship between

Table 1. Comparison of Demographic and Baseline Characteristics Between Patients with and without POD

Characteristic	With POD* (N = 24)	Without POD* (N = 550)	P Value
Demographics			
Age *(yr)*	74.5 ± 6.7	72.4 ± 5.7	0.114
BMI *(kg/m^2)*	27.6 ± 3.0	26.8 ± 3.7	0.287
Female sex	87.5%	90.5%	0.620
Use of glasses or hearing aid	13 (54.2%)	267 (48.5%)	0.495
Medical history			
Hypertension	10 (41.7%)	306 (55.6%)	0.165
Diabetes mellitus	6 (25.0%)	114 (20.7%)	0.628
Chronic kidney disease	1 (4.2%)	18 (3.3%)	0.816
Cerebrovascular disease	2 (8.3%)	33 (6.0%)	0.647
Solid tumor	3 (12.5%)	94 (17.1%)	0.547
History of delirium	2 (8.3%)	14 (2.6%)	0.084
Charlson Comorbidity Index	3.39 ± 2.21	1.96 ± 1.77	0.010
Socioeconomic factors			
Education: college or higher	4 (16.7%)	93 (16.9%)	0.740
Employment status: unemployed	20 (83.3%)	473 (86.0%)	0.775
Medical insurance coverage: insured	23 (95.8%)	537 (97.6%)	0.444
Presence of spouse	16 (66.7%)	386 (70.2%)	0.361
Presence of children	23 (95.8%)	529 (96.2%)	0.555
History of alcohol use	2 (8.3%)	46 (8.4%)	0.965
History of smoking	1 (4.2%)	16 (2.9%)	0.688
Laboratory data			
Hemoglobin *(g/dL)*	12.8 ± 1.2	12.9 ± 5.9	0.775
Blood urea nitrogen *(mg/dL)*	17.2 ± 15.5	19.0 ± 5.3	0.150
Creatinine *(mg/dL)*	0.78 ± 0.20	0.80 ± 0.34	0.642
Estimated glomerular filtration rate *(mL/min/1.73 m^2)*	79.0 ± 14.4	78.1 ± 13.7	0.772
Albumin *(g/dL)*	4.2 ± 0.3	4.3 ± 1.2	0.611
Sodium *(mEq/L)*	140.9 ± 2.2	140.1 ± 10.4	0.209
Potassium *(mEq/L)*	4.18 ± 0.86	4.42 ± 1.81	0.194
Erythrocyte sedimentation rate *(mm/hr)*	21.2 ± 5.1	19.8 ± 4.5	0.275
C-reactive protein *(mg/L)*	0.38 ± 0.09	0.32 ± 0.08	0.189
Preoperative medication usage			
Opioids	2 (8.3%)	39 (7.1%)	0.899
Sedatives	4 (16.7%)	36 (6.5%)	0.044
Antidepressants	3 (12.5%)	21 (3.8%)	0.027
Antipsychotics	1 (4.2%)	8 (1.5%)	0.236
Antiepileptics	1 (4.2%)	27 (4.9%)	0.642
Cholinergics	2 (8.3%)	33 (6.0%)	0.609
Anticholinergics	0 (0%)	1 (0.2%)	0.833
Hospital environment factors			
Illuminance *(lux)*	216.7 ± 24.9	222.3 ± 28.5	0.874
Noise level *(dB)*	50.5 ± 7.4	56.4 ± 15.2	0.002
Distance from window			0.940
First bed	9 (37.5%)	200 (36.4%)	
Second bed	7 (29.2%)	148 (26.9%)	
Third bed	8 (33.3%)	202 (36.7%)	
Surgical data			
ASA classification			0.089
1 or 2	20 (83.3%)	487 (88.5%)	
3 or 4	4 (16.7%)	63 (11.5%)	
Type of anesthesia: general	3 (12.5%)	21 (3.8%)	0.119
Intraoperative transfusion	1 (4.2%)	7 (1.3%)	0.151

*The values are given as the mean and the standard deviation, as the percentage of patients, or as the number of patients with the percentage in parentheses.

Table 2. Comparison of Preoperative Cognitive and Psychological Assessment Scores Between Patients with and without POD

	With POD* (N = 24)	Without POD* (N = 550)	P Value
CERAD battery			
J1, Verbal Fluency	10.5 ± 3.9	13.5 ± 3.4	0.002
J2, Boston Naming Test	6.6 ± 5.6	10.01 ± 4.7	0.005
J3, Mini-Mental State Examination	23.5 ± 3.2	26.3 ± 3.6	<0.001
J4, Wordlist Learning	9.0 ± 8.1	14.5 ± 8.0	0.002
J5, Constructional Praxis	5.7 ± 4.9	7.9 ± 4.2	0.035
J6, Wordlist Recall	2.9 ± 3.1	5.0 ± 3.0	0.004
J7, Wordlist Recognition	8.4 ± 2.5	7.9 ± 3.6	0.535
J8, Constructional Praxis Recall	7.9 ± 3.6	5.7 ± 3.7	0.007
J9, Trail Making Test Part A	78.3 ± 37.3	60.5 ± 33.7	0.130
J9, Trail Making Test Part B	192.0 ± 48.2	147.7 ± 39.6	0.102
CERAD total II (J1 + J2 + J4 + J5 + J6 + J7 + J8)	39.3 ± 30.2	51.7 ± 33.2	0.072
Psychological assessment scores			
PSQI	11.1 ± 4.2	8.4 ± 3.9	0.014
PHQ-15	5.9 ± 3.2	4.3 ± 2.6	0.014
HADS	8.3 ± 7.0	5.7 ± 5.4	0.104
SMCQ	3.8 ± 3.1	2.6 ± 2.0	0.095
SIRQD	2.6 ± 3.0	1.6 ± 5.2	0.461
Pain NRS	5.8 ± 1.1	5.5 ± 1.6	0.283

*The values are given as the mean and the standard deviation.

noise exposure and delirium risk likely depends on the measurement timing, clinical setting, and methodology, suggesting the need for future studies with more comprehensive environmental and clinical data[50].

By employing the complete CERAD neuropsychological battery, this study facilitated a more thorough assessment of cognitive function across multiple domains. The CERAD battery assesses multiple cognitive domains, such as verbal memory, executive function, visuospatial skills, and recognition memory, and is widely recognized for its efficacy in detecting cognitive impairments and Alzheimer disease[51,52]. Nonetheless, due to the high intercorrelation among CERAD subtests, it was not possible to include all variables in the final logistic regression model, to avoid multicollinearity. Previous studies have shown that the CERAD total score offers superior diagnostic accuracy over the MMSE in differentiating among normal aging, mild cognitive impairment, and Alzheimer disease[53,54]. However, in our study, the difference in the CERAD total II score between the group with POD and the group without POD did not reach significance in univariate analysis ($p = 0.072$). The SMCQ, which assesses subjective cognitive function from the patient perspective, and SIRQD, which assesses subjective cognitive function from the caregiver perspective, did not correlate with POD risk in our cohort. This indicates that objective cognitive assessments may offer stronger predictive value compared with subjective evaluations in the TKA population.

Our study identified higher PHQ-15 scores as an independent risk factor for POD. The PHQ-15 is a well-established tool for assessing somatic symptom burden, which commonly

Table 3. Regression Analysis of MMSE and Other Risk Factors for POD

Variable	OR*	Coefficient	P Value	VIF
Age in yr	1.001 (0.920 to 1.088)	0.010	0.994	1.312
BMI in kg/m^2	1.074 (0.947 to 1.218)	0.071	0.269	1.019
CCI	1.362 (1.035 to 1.792)	0.309	0.028	1.111
Noise level in dB	0.959 (0.912 to 1.008)	−0.042	0.100	1.059
Pain NRS	1.033 (0.771 to 1.384)	0.033	0.828	1.029
MMSE	0.771 (0.656 to 0.906)	−0.260	0.002	1.213

*The values are given as the OR, with the 95% CI in parentheses.

Table 4. Regression Analysis of PHQ-19 and Other Risk Factors for POD

Variable	OR*	Coefficient	P Value	VIF
Age in yr	1.033 (0.955 to 1.117)	0.032	0.421	1.125
BMI in kg/m^2	1.073 (0.960 to 1.200)	0.071	0.215	1.034
CCI	1.405 (1.062 to 1.859)	0.340	0.017	1.123
Noise level in dB	0.968 (0.924 to 1.013)	−0.033	0.156	1.074
Pain NRS	1.031 (0.769 to 1.383)	0.031	0.838	1.042
PHQ-15	1.187 (1.019 to 1.383)	0.172	0.028	1.060

*The values are given as the OR, with the 95% CI in parentheses.

accompanies psychological conditions such as depression and anxiety[55-57]. It is particularly effective in identifying psychosomatic symptoms that cannot be easily attributed to organic causes. Notably, the mean PHQ-15 score in our group with POD was 5.9, remaining below the commonly used clinical thresholds for somatic symptom burden, which include 6 or 7 for *DSM-IV (Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition)* somatoform disorders and 9 in occupational health settings[58,59]. Given the moderate to good diagnostic accuracy of the PHQ-15 in detecting psychosomatic distress in the general population, our findings highlight the importance of systematic preoperative screening for even mild somatic symptoms, especially in patients undergoing elective TKA[60].

This study had several limitations. First, although the study was executed at 2 major tertiary care hospitals with similar hospitalization protocols and surgical procedures for patients who underwent TKA, these findings may have limited generalizability to institutions with different perioperative management strategies. In particular, because all patients were admitted the day prior to surgery, variations in admission practices and duration across institutions may further restrict the applicability of our results to other clinical settings. Second, the relatively small sample size of the group with POD may have reduced statistical power, potentially limiting the ability to detect other contributing factors. Although a post hoc power analysis indicated sufficient power to detect correlations with an OR of 3.0, the ability to identify smaller effects may have been limited. Third, psychological assessments, including the PSQI and PHQ-15, relied on self-reported measures, which could be susceptible to response bias. More objective measures such as actigraphy for sleep monitoring or physiological stress markers may provide a more comprehensive understanding of their role in POD risk. Lastly, environmental factors, including noise and illuminance levels, were assessed on a single preoperative day and may not have fully captured the dynamic nature of hospital environments during the perioperative period.

In conclusion, this prospective study comprehensively evaluated preoperative cognitive function and psychological symptoms in patients undergoing TKA. Lower preoperative MMSE scores and higher PHQ-15 scores were correlated with an increased risk of POD, even at subclinical levels. These findings underscore the importance of preoperative screening to identify patients at high risk for POD.

Appendix

eA Supporting material provided by the authors is posted with the online version of this article as a data supplement at jbjs.org (http://links.lww.com/JBJS/J18). ■

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